Winning baseball games by solving statistics puzzles

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ICSDS 2025

RICE UNIVERSITY
Sport Analytics

Outline

1. Pickoff Game Theory
   - Linear mixed-effects model (LMM)
   - Markov decision process (MDP)
2. Predictive Pitch Score
   - Bayesian hierarchical model
   - Supervised learning
3. Swinging, Fast and Slow
   - Bayesian hierarchical model (skew-normal!)
   - Instrumental variables regression

Part I: Pickoff Game Theory

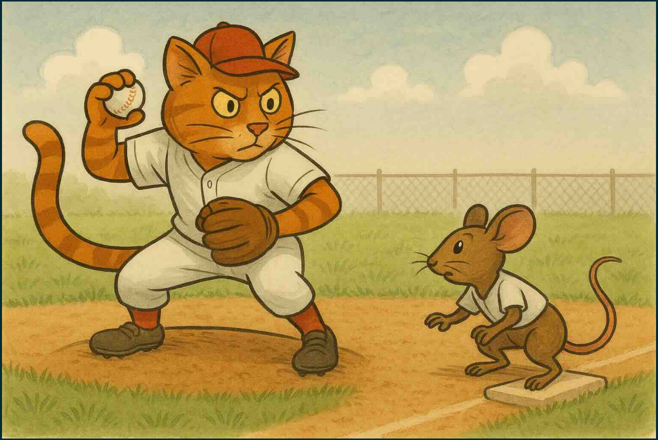
 [Jacob Hahn](), Rice University → Chicago Cubs

 Mallesh Pai, Rice University

 Andrew Schaefer, Rice University

Working paper:
Hahn J, Powers S, Pai M, Schaefer A "The Two-Foot Rule: A game theoretic analysis of the pickoff limit in Major League Baseball"

Background



- In 2023, MLB limited the pitcher to 3 pickoff attempts
- After 3 failed pickoffs, the runner advances automatically

The Puzzle

If you are the runner, how much should you extend your lead after each pickoff attempt?

Data: pitch-by-pitch lead distance (player tracking) from MLB*

Our Approach

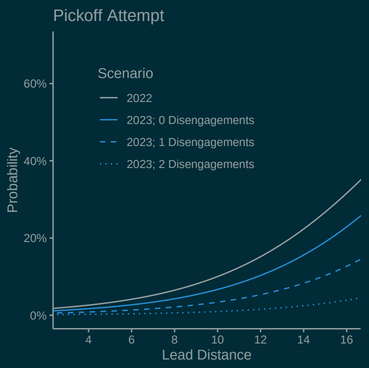
Step 1: Estimate the effect of lead distance

- Linear mixed-effects models (LMMs) for probabilities of pickoff attempt, pickoff success, stolen base success
- Fixed effects: lead distance, disengagements, outs, count, catcher arm strength, runner sprint speed
- Random effects: pitcher, catcher, runner identities

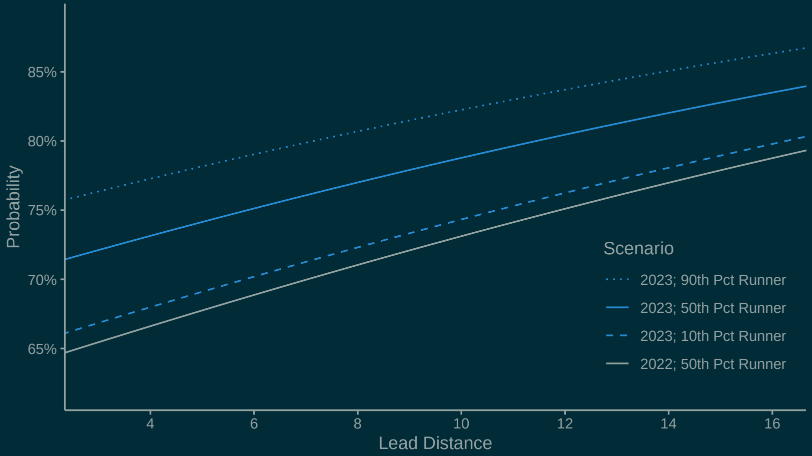
Step 2: Solve for the optimal lead distance

- Markov decision process (MDP) to model an inning
- State space: disengagements, bases occupied, outs, count
- Action space: lead distance
- Reward function: runs scored

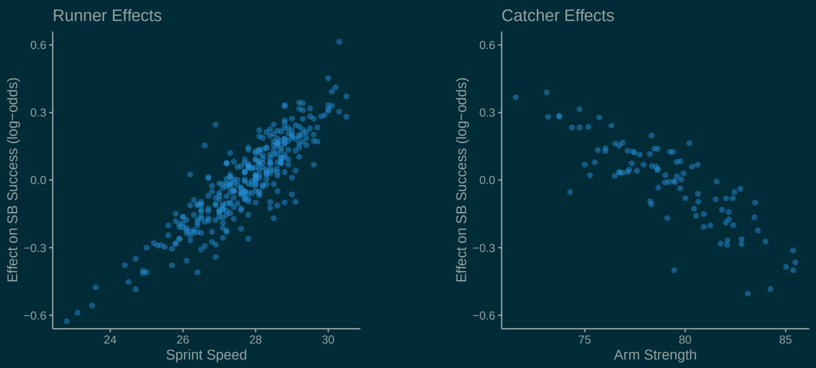
Results: Pickoff Attempt and Success Probability



Results: Stolen Base Success Probability



Results: Runner and Catcher Effects



Results: Optimal Lead Distance

	Average Lead		Actual Exceeds
Disengagements	Actual	Recommended	Recommendation
0	9.5	10.3	31.1%
1	10.2	11.7	12.4%
2	10.8	14.1	1.0%

Additional results in paper:

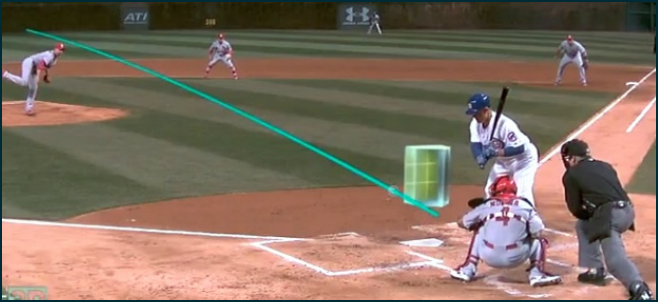
- What if you are not an average runner?
- What if the pitcher has agency? (game theory)

Part II: Predictive Pitch Score

Vicente Iglesias, Susquehanna International Group

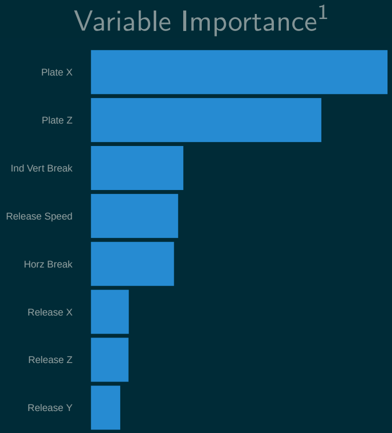
Working paper:
Powers S, Iglesias V "Pitch trajectory density estimation for predicting future outcomes"

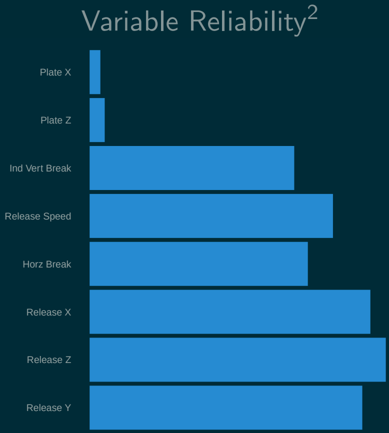
Background



- Full pitch tracking data have been available for two decades
- Trajectory estimated as 3D quadratic function of time ($\in \mathbb{R}^9$)
- Standard practice: Evaluate pitchers based on trajectories

The Puzzle





[1] fractional contribution of each feature's splits to gradient boosting pitch model
[2] (between-pitcher variance) / (total variance); varies by pitch type (here: RHB FB)

Why Supervised Learning Isn't Enough

Supervised Learning

Pitch		Outcome
x_1	$\rightarrow$	y_1
x_2	$\rightarrow$	y_2
x_3	$\rightarrow$	y_3
	$\cdots$	
x_n	$\rightarrow$	y_n
x^*	$\rightarrow$	$\hat{y}$

Our Problem

Pitcher	Pitch		Outcome
A	x_1	$\rightarrow$	y_1
A	x_2	$\rightarrow$	y_2
B	x_3	$\rightarrow$	y_3
		$\cdots$	
C	x_n	$\rightarrow$	y_n
		$\searrow$	
A			$\hat{y}$

Why Supervised Learning Isn't Enough

Supervised Learning

Pitch		Outcome
x_1	$\rightarrow$	y_1
x_2	$\rightarrow$	y_2
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	$\dots$	
x_n	$\rightarrow$	y_n
x^*	$\rightarrow$	$\hat{y}$

Our Approach

Pitcher	Pitch		Outcome
A	x_1	$\rightarrow$	y_1
A	x_2	$\rightarrow$	y_2
B	x_3	$\rightarrow$	y_3
		$\dots$	
C	x_n	$\rightarrow$	y_n
	$\downarrow$		
A	$\hat{p}(x)$	$\rightarrow$	$\int \hat{p}(x)\hat{f}(x)$

Our Approach

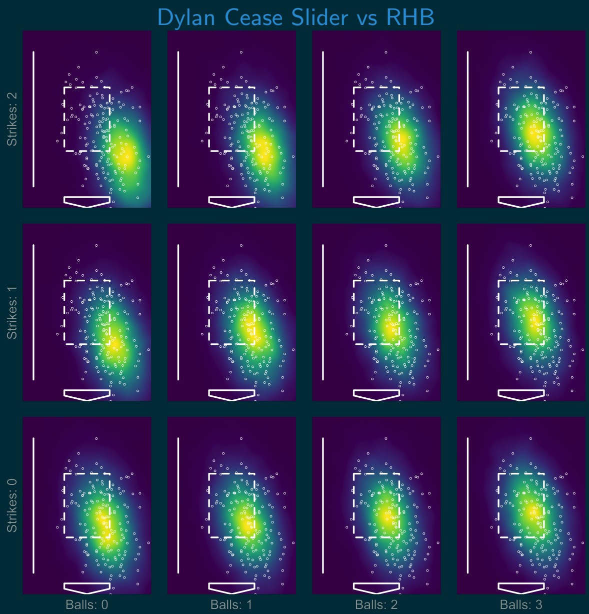
Step 1: Estimate trajectory distributions conditional on covariates

- Bayesian hierarchical model (9D MVN likelihood)
- Mean $\sim$ pitcher, context, pitcher $\times$ context
- Variance scaling $\sim$ pitcher, context
- Correlation $\sim$ pitcher
- Fit via maximum *a posteriori* (MAP) estimator

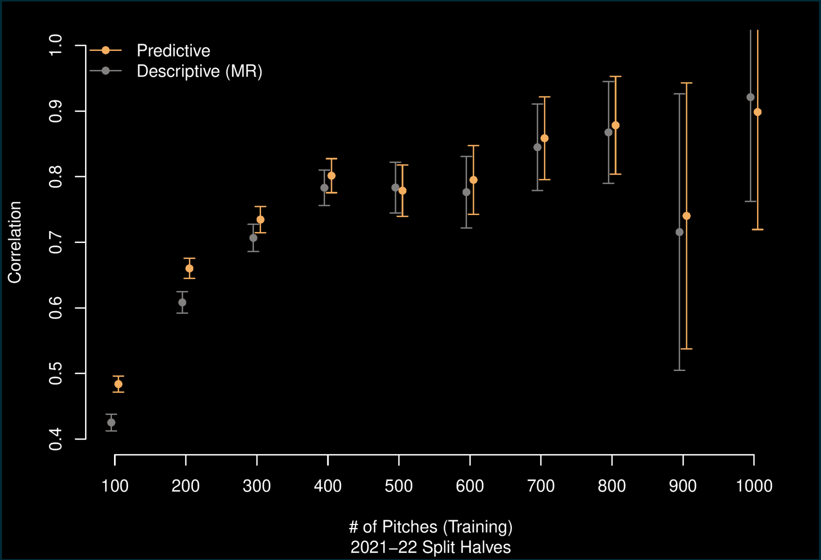
Step 2: Fit supervised learning model and integrate w.r.t. Step 1

- Gradient boosting

Results: Illustration of Estimated Distribution



Results: Predictive Performance



Part III: Swinging, Fast and Slow

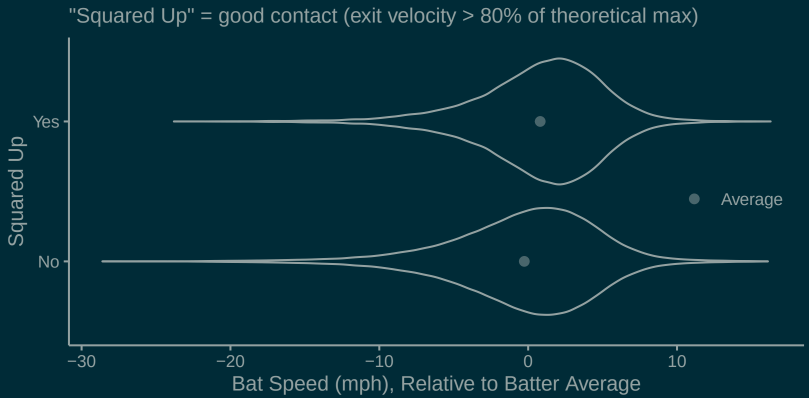
Ron Yurko, Carnegie Mellon University

Working paper:
Powers S, Yurko R "Swinging, Fast and Slow: Interpreting variation in baseball swing tracking metrics"

Background

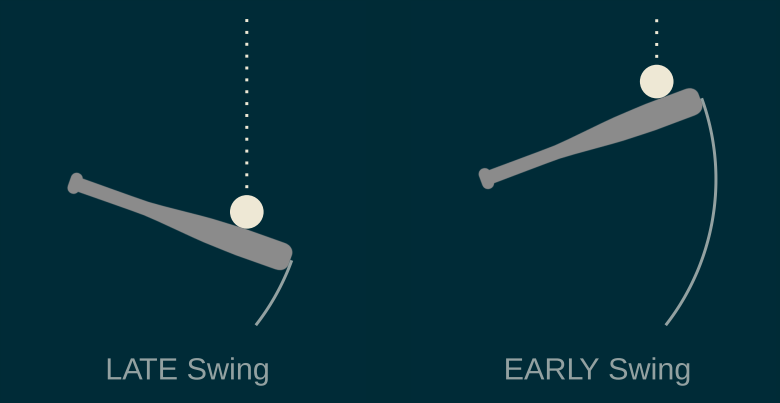
- In May 2024, MLB released swing length and bat speed data
- Both are measured at contact (or at point nearest to contact)

The Puzzle



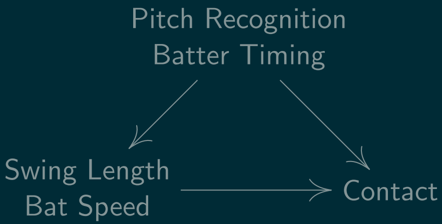
- How should batters modulate swing length and bat speed?

Challenge #1: Measurement



- For identical swings, timing determines point of measurement
- Swing length and bat speed are outcomes, not purely process

Challenge #2: Confounding



Our Approach

Step 1: Estimate batter intention

- Bayesian hierarchical model (skew-normal likelihood)
- Condition on successful contact against fastballs!
- Mean $\sim$ batter, count, location, batter $\times$ (count, location)
- Skew $\sim$ batter

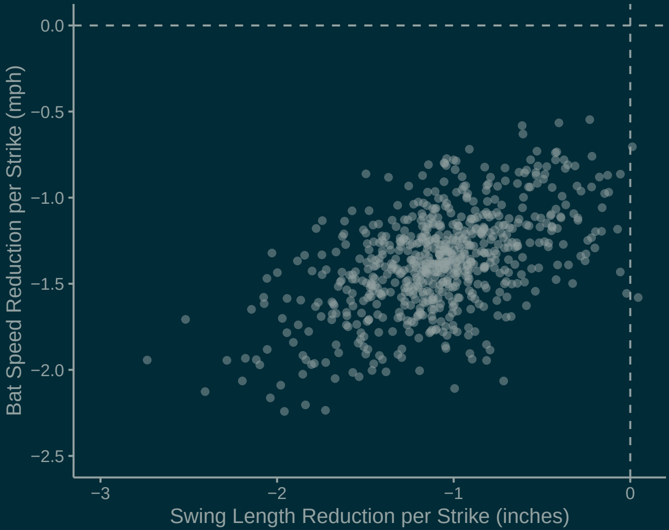
Step 2: Estimate effect of batter intention

- Instrumental variable regression (instrument: count)
- Control for pitch difficulty using pitch outcome model (Part II)
- Response: contact (logistic reg.) and "power" (linear reg.)

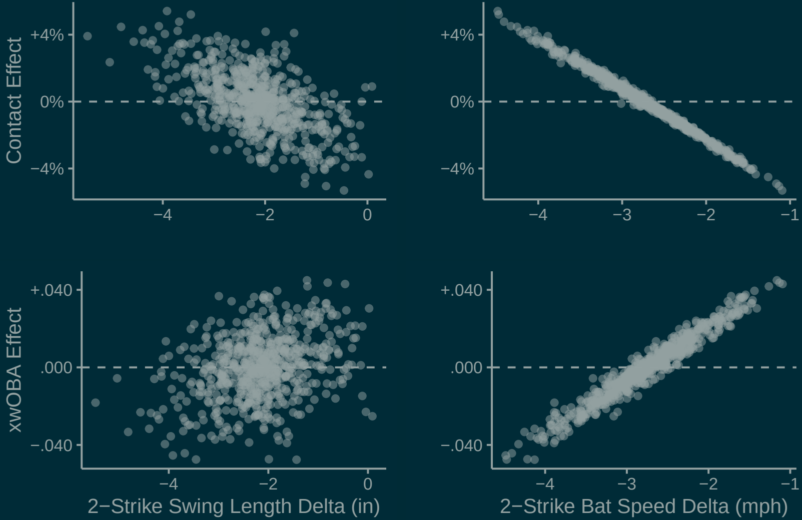
Step 3: Value the tradeoff between contact and power

- Model the plate appearance as a Markov chain

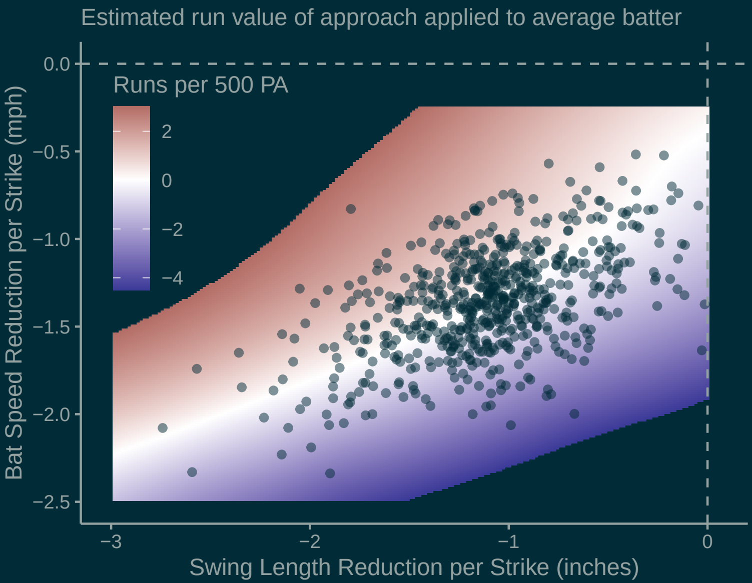
Results: Batter × Count Interaction



Results: Causal Effect Estimates



Results: Contact-Power Tradeoff



Thank You!

saberpowers.github.io